Dark Matter as Khronon Condensate: Weak-Field Phenomenology of the τ Framework

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The temporal asymmetry parameter $\tau = 1 - F$, introduced in Paper 1 as the Petz recovery failure, acquires physical realization through the Blanchet–Skordis (BS) Khronon action $S = \int \sqrt{-g}\,[R - 2J(Y) + 2K(Q)]\,d^4x$. This action contains two structurally independent sectors: the *cosmological sector* $K(Q) = \mu^2(Q-1)^2$, fixed by ghost condensation with a single parameter $\mu_0 = H_0/c$ from dimensional analysis, which reproduces the cosmological dark matter density $\Omega_{\rm DM} = \delta_0(2+\delta_0)/3 = 0.265$ with CDM-like perturbations ($c_{\rm s}^2 = 0$); and the *galactic sector* $J(Y)$, a phenomenological free function constrained only at its asymptotic limits, which reproduces the Baryonic Tully–Fisher Relation and the Radial Acceleration Relation with one empirical parameter $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. We prove a no-go theorem establishing that pure scalar Khronon dark matter is incompatible with CMB observations, necessitating the Blanchet–Skordis tensor field. The resulting Khronon-Tensor theory predicts a CMB power spectrum *identical* to ΛCDM — consistent with Planck data — while producing MOND dynamics at galaxy scales. We present a systematic confrontation of this two-sector framework with 11 low-redshift observational tests (10 passed, 1 marginal), show that the Bullet Cluster mass-gas offset is naturally explained by the $c_{\rm s}^2 = 0$ collisionless Khronon condensate, compare its parameter economy with ΛCDM and AeST, and identify falsifiable predictions. We are explicit about what is derived, what is assumed, and what remains open—in particular, the connection between a_0 and μ_0 is suggestive but unproven, and the running $\mu(k) = k$ is a well-motivated assumption, not a derivation.

I. INTRODUCTION

The dark matter problem encompasses two distinct observational domains: (i) cosmological—the matter density inferred from the cosmic microwave background (CMB), baryon acoustic oscillations (BAO), and large-scale structure; and (ii) galactic—the mass discrepancy observed in galaxy rotation curves, the Baryonic Tully–Fisher Relation (BTFR), and the Radial Acceleration Relation (RAR). The standard ΛCDM model addresses both with a single ingredient—cold dark matter (CDM) particles—but despite four decades of searches, no particle has been directly detected [? ? ?].

Modified gravity approaches (MOND [?], TeVeS [?], AeST [? ?]) excel at galactic phenomenology but have historically struggled with cosmology. The breakthrough of Skordis and Zlósnik [?] demonstrated that a relativistic MOND theory *can* reproduce both the CMB power spectrum and the matter power spectrum, provided the theory contains a scalar field whose perturbations mimic CDM at early times ($c_{\rm s}^2 = 0$).

In Paper 1 [?], we introduced the temporal asymmetry parameter $\tau = 1 - F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$, which quantifies the arrow of time as Petz recovery failure, with the universal bound $\tau \leq 1 - \exp(-\Sigma/2)$. In Paper 2 [?], we showed that in strong gravitational fields $\Sigma_{\rm grav} = -\ln(-g_{00})$, connecting the Petz recovery fidelity to the gravitational channel transmissivity $\eta = -g_{00}$.

This paper addresses the *weak-field* phenomenology of the same framework. We adopt the Blanchet–Skordis (BS) Khronon action [?] as the concrete realization of the τ framework at galactic and cosmological scales. Our goals are:

1. To present the BS Khronon theory with honest parameter counting and epistemic transparency;

2. To show that $K(Q) = \mu^2(Q-1)^2$ with $\mu_0 = H_0/c$ reproduces cosmological dark matter naturally;

3. To show that $J(Y)$ reproduces MOND galactic phenomenology;

4. To systematically confront the theory with low-redshift observations;

5. To identify what is derived, what is assumed, and what remains open.

What this paper does not claim.—We do not claim to "solve" the dark matter problem. We provide a new interpretation in which dark matter phenomenology emerges from the ghost condensation of a Khronon field—the field that defines the observer's time direction in the τ framework. The theory contains a free function $J(Y)$ (the MOND interpolating function), and two key assumptions (running μ, the a_0–μ_0 coincidence) remain unproven.

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II. THE BLANCHET–SKORDIS KHRONON ACTION

A. Action and field content

[KNOWN] The BS Khronon theory [? ?] describes gravity coupled to a single scalar field ϕ (the "Khronon") whose level surfaces define spatial hypersurfaces. The action is

$$S = \frac{c^3}{16\pi G}\int d^4x\,\sqrt{-g}\,\big[R - 2J(Y) + 2K(Q)\big] + S_{\rm m}, \quad (1)$$

where $S_{\rm m}$ is the matter action (universally coupled to $g_{\mu\nu}$ only), and:

- The *unit normal* to the $\phi = \text{const}$ hypersurfaces is $n_\mu = -(c/Q)\,\nabla_\mu\phi$, with $Q = c\sqrt{-g^{\mu\nu}\nabla_\mu\phi\,\nabla_\nu\phi}$.
- The *acceleration* of the foliation is $A_\mu = c^2\,n^\nu\nabla_\nu n_\mu$, with $Y = A_\mu A^\mu/c^4$.
- $K(Q)$ controls the **cosmological sector**: the energy density and equation of state of the Khronon field on the FRW background.
- $J(Y)$ controls the **galactic sector**: the quasi-static, nonrelativistic limit that reproduces MOND phenomenology.

[KNOWN] The Khronon field is *hypersurface-orthogonal by construction* (n_μ is a gradient), which eliminates the twist tensor $\omega_{\mu\nu} = 0$ and merges the Einstein–Aether parameters c_1 and c_3 into the single combination $c_{13} = c_1 + c_3$ [? ? ?].

B. Why $c_{13} = 0$: action choice, not geometry

[ASSUMED] The BS action (??) does *not* contain the extrinsic curvature coupling $K_{\mu\nu}K^{\mu\nu}$ (where $K_{\mu\nu}$ is the extrinsic curvature of the $\phi = \text{const}$ hypersurfaces). In the Einstein–Aether parametrization, this coupling generates $c_{13} \neq 0$. Its absence yields $c_{13} = 0$ *by construction* [?], giving the gravitational wave speed

$$c_{\rm T}^2 = \frac{1}{1 - c_{13}} = 1, \quad \Longrightarrow \quad c_{\rm T} = c \ \text{(exactly)}. \quad (2)$$

This is a *choice of action*, not a consequence of hypersurface orthogonality alone (which only merges c_1, c_3 into c_{13} but does not set it to zero). The choice is strongly motivated by:

1. GW170817 [?]: $|c_{\rm T}/c - 1| < 10^{-15}$;
2. Theoretical minimality (fewer terms in the action);
3. Even if a $K_{\mu\nu}K^{\mu\nu}$ term were added, the tiny coupling $\mu_0 = H_0/c$ would make $c_{13} \sim 10^{-30}$ at solar system scales [?].

C. Degrees of freedom and stability

[KNOWN] The hypersurface-orthogonal Khronon propagates 3 degrees of freedom [?]: 2 tensor (spin-2 graviton) + 1 scalar (Khronon mode). Vector modes are eliminated by the constraint $\omega_{\mu\nu} = 0$.

For the kinetic function $K(Q)$ with a minimum at $Q = Q_0$, the scalar sector is free of all standard instabilities [? ?]:

- **No Ostrogradski ghost**: the field equation is second-order in ϕ (BPS [?], Hamiltonian analysis);
- **No wrong-sign kinetic term**: $K''(Q_0) > 0$ ensures positive-definite kinetic energy;
- **No gradient instability**: $c_{\rm s}^2 = K'(Q_0)/[K'(Q_0) + 2Q_0\,K''(Q_0)] = 0$ (marginal, cured by ghost condensation k^4 terms [?]);
- **No tachyon**: Q_0 is a global minimum of K;
- $c_{\rm T} = c$ **exactly**: $c_{13} = 0$ by construction;
- **Strong coupling at** $\Lambda_3 \sim (\mu_0^2 M_{\rm Pl})^{1/3} \sim 10^{-13}$ **eV**: far above all cosmological and astrophysical scales ($H_0 \sim 10^{-33}$ eV).

The full 8-criterion stability analysis is presented in the Supplemental Material [?].

D. Honest parameter counting

The theory contains:

- 6 standard cosmological parameters (H_0, $\Omega_{\rm b}h^2$, n_s, A_s, $\tau_{\rm reion}$, Σm_ν);
- 1 cosmological parameter: μ (Khronon mass, or equivalently $\Omega_{\rm DM}h^2$ as an initial condition);
- 1 galactic parameter: a_0 (MOND acceleration scale);
- 1 free function: $J(Y)$ (MOND interpolating function, constrained only at asymptotic limits).

TABLE I. Parameter counting comparison. "func" denotes a free function. $\Omega_{\rm DM}h^2$ is an initial condition (fitted, not predicted) in all theories, including ΛCDM.

Theory	Cosmo	Theory-specific	Free func.
ΛCDM	6	$\Omega_{\rm DM}h^2$	0
MOND (non-rel.)	—	a_0	$\mu(x)$
AeST [?]	6	$K_B, K_2, \lambda_s, Q_0, a_0$	$F(Y,Q)$
BS Khronon	6	$\mu_0, a_0, \Omega_{\rm DM}h^2$	$J(Y)$
BS Khronon + τ	6	$a_0, \Omega_{\rm DM}h^2$	$J(Y)$

Table **??** compares parameter counts across theories. The τ framework fixes $\mu_0 = H_0/c$ via dimensional analysis (Sec. **??**), reducing the theory-specific parameter count by one relative to plain BS Khronon. The free function $J(Y)$ is inherited from the MOND tradition dating to Bekenstein and Milgrom [**?**], and has been free for forty years. The Crooks interpolation function $\mu(x) = 1 - e^{-\sqrt{x}}$ (Sec. **??**) could in principle eliminate this freedom if validated against SPARC data.

III. COSMOLOGICAL SECTOR: $K(Q)$

A. Ghost condensation fixes $K(Q)$

[DERIVED] Any scalar field that spontaneously breaks Lorentz invariance by acquiring a timelike gradient $\langle \partial_\mu \phi \rangle \neq 0$ sits in the "ghost condensation" regime [**?**]. The leading-order effective action around the condensate $Q = 1$ is

$$K(Q) = \mu^2 \, (Q-1)^2, \tag{3}$$

as established by Arkani-Hamed et al. [**?**] and Scherrer [**?**]. The quadratic form is not arbitrary: it is the unique leading-order expansion around a kinetic minimum with $K'(1) = 0$ and $K''(1) = 2\mu^2 > 0$, ensuring stability and $c_s^2 = 0$. Petz optimality provides an independent argument: the maximally recoverable channel configuration corresponds to a quadratic kinetic function.

Higher-order terms $K(Q) = \mu^2(Q-1)^2 + \alpha_3(Q-1)^3 + \cdots$ are possible. The DBI completion [**?**] $K_{\rm DBI} = \mu^2\lambda_D^2[\sqrt{1+(Q-1)^2/\lambda_D^2} - 1]$ provides a natural UV regularization, introducing at most one additional parameter λ_D. We focus on the leading quadratic form throughout.

B. $\mu_0 = H_0/c$: dimensional analysis

[ASSUMED] The coupling μ has dimensions $[\mathrm{m}^{-1}]$. The available fundamental constants are $\{H_0, c, \hbar, G, k_{\rm B}\}$. The general combination with these dimensions is [**?**]

$$\mu = \frac{H_0}{c} \, (t_{\rm P} H_0)^{2d}, \tag{4}$$

where $t_{\rm P} = \sqrt{\hbar G/c^5}$ is the Planck time and $t_{\rm P} H_0 = 1.18 \times 10^{-61}$. For *any* $d \neq 0$, the correction factor is $10^{-122|d|}$, destroying the result. Therefore

$$\boxed{\mu_0 = \frac{H_0}{c} = 7.28 \times 10^{-27} \ \mathrm{m}^{-1}} \tag{5}$$

is the *unique* selection without Planck-scale physics. This is a naturality argument, not a derivation from field equations. Three equivalent physical interpretations hold:

1. Khronon Compton wavelength $= c/H_0 =$ Hubble radius;
2. $\mu_0 = 2\pi k_{\rm B} T_{\rm dS}/(\hbar c)$: inverse de Sitter thermal wavelength [**?**];
3. $\mu_0 = 2\pi a_0/c^2$: connects to the MOND scale (Sec. **??**).

C. Cosmological background: FRW solution

[KNOWN] On the FRW background, the Khronon field equation gives [**?**]

$$K'(Q_0) = \frac{I_0}{a^3 Q_0}, \tag{6}$$

where I_0 is an integration constant (the initial Khronon momentum). This conservation law holds even when μ varies in time, since the Lagrangian has no explicit ϕ-dependence ($\partial\mathcal{L}/\partial\phi = 0$). Substituting $K'(Q_0) = 2\mu^2\delta$ gives

$$Q_0(a) = 1 + \delta(a), \qquad \delta(a) = \frac{I_0}{2\mu^2(a)\, a^3}, \tag{7}$$

where $\mu(a)$ is the (possibly running) coupling.

A crucial self-consistency: regardless of how μ runs, the combination $\mu^2\delta = I_0/(2a^3)$ depends only on a. Therefore the Khronon energy density (**??**) satisfies $\rho_K \propto (2+\delta)/a^3$, which is CDM-like ($\rho_K \propto a^{-3}$) whenever $(2+\delta)$ varies slowly—i.e., whenever $\delta \lesssim O(1)$.

For constant $\mu = \mu_0$: $\delta \propto a^{-3}$, giving $\delta(z = 1100) \sim 10^9 \gg 1$—the Khronon exits the ghost condensation regime and the energy density scales as $\rho_K \propto a^{-6}$ (stiff matter), incompatible with CMB observations. This is the true **CMB Catch-22** (Sec. **??**). Running $\mu_{\rm bg} = H(a)/c$ keeps $\delta \lesssim O(1)$ at all epochs (Sec. **??**), maintaining the ghost condensation regime and CDM-like dilution.

The Khronon energy density follows from the k-essence stress-energy $\rho = QK' - K$ [**?**]:

$$\rho_K = \frac{c^2\mu^2\,\delta\,(2+\delta)}{8\pi G}, \tag{8}$$

where the factor $(2+\delta)$ arises from $QK' - K = (1+\delta)\cdot 2\mu^2\delta - \mu^2\delta^2 = \mu^2\delta(2+\delta)$. For $\delta \ll 1$, $\rho_K \approx c^2\mu^2\delta/(4\pi G)$; for $\delta \gg 1$, $\rho_K \sim c^2\mu^2\delta^2/(8\pi G)$. Since $\mu^2\delta = I_0/(2a^3)$, one has $\rho_K = c^2 I_0\,(2+\delta)/(16\pi G\, a^3)$: the energy density dilutes as a^{-3} (CDM-like) with $O(1)$ corrections encoded in $(2+\delta(a))$.

The equation of state follows from $w = K/\rho_K = \mu^2\delta^2/[\mu^2\delta(2+\delta)]$ [**?**]:

$$w = \frac{\delta}{2+\delta}. \tag{9}$$

Importantly, w does *not* map directly to the GDM equation of state $w_{\rm GDM}$ constrained by Thomas, Kopp & Skordis [**?**]. The actual instantaneous dilution rate is $d\ln\rho_K/d\ln a = -3 - d\ln(2+\delta)/d\ln a$, which equals -3 whenever δ varies slowly.

Theorem 1 (Khronon Information Conservation)**.** *For the Blanchet–Skordis action with $K(Q) = \mu^2(Q-1)^2$ on FRW background, the scalar field equation implies*

$$a^3\, Q_0\, K'(Q_0) \;=\; I_0 \;=\; \text{const}, \tag{10}$$

where I_0 is the conserved Khronon momentum. This holds for arbitrary time-dependent $\mu(t)$.

Proof sketch. The Lagrangian density $\mathcal{L}(\partial\phi, \partial^2\phi)$ has no explicit ϕ-dependence ($\partial\mathcal{L}/\partial\phi = 0$). By the Euler–Lagrange equation, the canonical current $J^a = \partial\mathcal{L}/\partial(\partial_a\phi)$ satisfies $\partial_a(\sqrt{-g}\, J^a) = 0$. On FRW with $\phi = t + \delta\phi(t)$, the spatial components vanish by isotropy, leaving the temporal conservation law (**??**). □

Consequences:

1. **CDM-like dilution identity**: $\mu^2\delta = I_0/(2a^3)$—the product is independent of how μ runs.
2. **Running μ stability**: with $\mu = H(a)/c$, $\delta(a) = I_0 c^2/(2H^2 a^3)$ stays $O(1)$ at all epochs.
3. **Automatic a^{-3} scaling**: $\rho_K = c^2 I_0\,(2{+}\delta)/(16\pi G\, a^3)$ dilutes as dust whenever $(2{+}\delta) \approx$ const.

Remark (Unified entropy production).—On static backgrounds $Q = 1/\sqrt{-g_{00}}$, giving $\Sigma_{\text{grav}} = -\ln(-g_{00}) = 2\ln Q$ (Paper 2 [**?**]). On FRW backgrounds $Q_0 = 1+\delta$, giving $\Sigma_{\text{cosmo}} = 2\ln(1+\delta)$. Both are instances of the general result $\Sigma = 2\ln Q$, where Q is the inverse Khronon lapse. This identity holds exactly on stationary and FRW backgrounds, and to $O(\varepsilon_{\text{NA}}^2)$ on any adiabatic background, where $\varepsilon_{\text{NA}} = |d\ln Q/d\tau|/(2\omega)$ is the non-adiabaticity parameter of the probe mode. The correction is always non-negative (particle creation adds entropy), so $\Sigma \geq 2\ln Q$ in general.

D. Running μ: assumption and motivation

[ASSUMED] With constant $\mu = \mu_0 = H_0/c$, $\delta(a) = \delta_0/a^3$ grows as a^{-3}, reaching $\delta \sim 10^9$ at $z = 1100$. In this regime the Khronon exits ghost condensation: $(2{+}\delta) \sim \delta$, so $\rho_K \propto \delta^2/a^3 \propto a^{-9}$ (stiff matter, $w \to 1$). This is the **CMB Catch-22**: constant $\mu_0 = H_0/c$ drives $\delta \gg 1$ at early times, destroying CDM-like behavior.

We *assume* the running

$$\mu_{\text{bg}}(a) = \frac{H(a)}{c}, \tag{11}$$

motivated by three convergent arguments:

1. **DPI + extensivity**: the anomalous dimension $\eta_\mu = d - 3 = 1$ in $d = 4$ follows from the transition from area-law to volume-law entanglement entropy at the de Sitter scale, combined with the data processing inequality [**?** **?** **?** **?**];
2. **Dimensional transmutation**: $\mu(k) = k$ is the unique power-law running independent of the boundary condition H_0; Kumar [**?**] obtains a logarithmic correction to Newton's potential from marginal IR running in QFT, providing independent support;
3. **Modular flow**: if the Khronon is the modular flow direction, the modular temperature $T_{\text{mod}} = k/\pi$ gives $\mu(k) \sim k$.

A rigorous derivation from the microscopic theory remains an open problem.

With (**??**), $\delta(a) = I_0 c^2/(2H^2(a)\, a^3)$, which gives:

- **Radiation era** ($H^2 \propto a^{-4}$): $\delta \propto a \to 0$ at early times;
- **Matter era** ($H^2 \propto a^{-3}$): $\delta \approx \delta_0/\Omega_m \sim O(1)$;
- **Today**: $\delta_0 = 0.340$, $w_0 = \delta_0/(2{+}\delta_0) = 0.145$.

The running keeps $\delta \lesssim O(1)$ at all epochs, maintaining the ghost condensation regime. Since $\mu^2\delta = I_0/(2a^3)$ is independent of the running, the Khronon energy density $\rho_K \propto (2{+}\delta)/a^3$ remains CDM-like, with $O(1)$ corrections during the matter era that require a full Boltzmann solver to quantify. The perturbation sound speed $c_{\text{s}}^2 = 0$ is unaffected.

E. Perturbations: $c_{\text{s}}^2 = 0$ (CDM-like)

[DERIVED] The scalar sound speed is [**?**]

$$c_{\text{s}}^2 = \frac{K'(Q_0)}{K'(Q_0) + 2Q_0\, K''(Q_0)} \xrightarrow{K'(1)=0} 0. \tag{12}$$

This is not fine-tuned: it follows from $K(Q)$ having a *minimum* at $Q_0 = 1$ (ghost condensation). The $c_{\text{s}}^2 = 0$ condition is radiatively stable, with quantum corrections $\delta c_{\text{s}}^2 \sim (\mu/M_{\text{Pl}})^2 \sim 10^{-122}$ [**?**].

Consequence: Khronon perturbations grow identically to CDM at all cosmologically relevant scales. The Jeans length $\lambda_J = 0$ (no pressure support), so structure formation proceeds as in ΛCDM. In particular, the CMB acoustic peak structure, matter power spectrum $P(k)$, and BAO features are reproduced without modification.

Physical meaning of $c_{\text{s}}^2 = 0$.—The vanishing sound speed means Khronon perturbations do *not* propagate: they grow in place, exactly like CDM density perturbations. By Theorem **??**, $K'(Q_0) = I_0/(a^3 Q_0) \to 0$ as the condensate approaches $Q_0 = 1$; since the numerator in (**??**) is $K'(Q_0)$ while the denominator retains $2Q_0K''(Q_0) = 4\mu^2 > 0$, the ratio vanishes identically at the ghost condensation point. This resolves the apparent tension with the GDM constraints of Thomas, Kopp & Skordis [**?**]: the BS effective equation of state $w = \delta/(2{+}\delta)$ (**??**) does *not* equal the GDM sound speed $c_{\text{s,GDM}}^2$. The two quantities are structurally

independent—w governs the background dilution rate while c_s^2 is set by the kinetic structure of $K(Q)$. Since $c_s^2 = 0$ exactly, the TKS bound $c_s^2 < 3.21 \times 10^{-6}$ (99.7% CL) is trivially satisfied.

F. Cosmological predictions

With $\mu_0 = H_0/c$ and running (**??**):

- $\Omega_{\rm DM}$ **prediction**: Since $c^2\mu_0^2/(8\pi G) = H_0^2/(8\pi G) = \rho_{\rm crit}/3$, Eq. (**??**) gives

$$\Omega_{\rm DM} = \frac{\rho_K}{\rho_{\rm crit}} = \frac{\delta_0(2+\delta_0)}{3}. \tag{13}$$

  For $\delta_0 = 0.34$: $\Omega_{\rm DM} = 0.34 \times 2.34/3 = 0.265$, in agreement with observations [**?**];

- $w(z{=}0) = \delta_0/(2{+}\delta_0) = 0.145$ (for $\delta_0 = 0.340$): a specific prediction for the late-time dark matter equation of state, testable by DESI DR2 (2027) and Euclid (2028);

- Linear $P(k)$ identical to CDM; nonlinear deviations at $k > 1\ h$/Mpc are a prediction [**?**];

- $a_0 = c^2\mu_0/(2\pi) = cH_0/(2\pi) = 1.04 \times 10^{-10}$ m/s^2 (for $H_0 = 67.4$ km/s/Mpc; within 13% of empirical $a_0 \approx 1.2 \times 10^{-10}$ m/s^2).

Note that $\Omega_{\rm DM}h^2 = 0.12$ is an *initial condition* (the value of I_0), not a prediction of the theory—as it is in ΛCDM, where $\Omega_{\rm DM}h^2$ is likewise a fitted parameter.

G. CMB caveat: the $(2{+}\delta)$ correction

We note that the factor $(2{+}\delta)$ in Eq. (**??**) introduces a redshift-dependent correction to the dark matter density: $\rho_K(z)/\rho_{\rm CDM}(z) = (2{+}\delta(z))/(2{+}\delta_0)$, where δ varies from ~ 0 (radiation era) to $\delta_0/\Omega_m \approx 1.08$ (matter era) to $\delta_0 = 0.34$ (today). This $\sim$20% variation at recombination is potentially constrainable by CMB observations and represents an important test of the framework. A DBI completion of $K(Q)$ at large δ may regularize this correction (see Discussion).

IV. GALACTIC SECTOR: $J(Y)$

A. $J(Y)$ is a free function

[KNOWN] In the quasi-static, weak-field limit ($Q \to 1$, $K \to 0$), only the $J(Y)$ sector of the action contributes to galactic dynamics [**?** **?**]. Conversely, in the cosmological limit ($Y \to 0$, $J \to \Lambda$), only $K(Q)$ matters. **The two sectors decouple.**

The function $J(Y)$ is constrained only at its asymptotic limits [**?**]:

$$\text{Deep-MOND } (Y \ll a_0^2/c^4): \quad J(Y) \sim \Lambda - Y + \frac{c^2}{a_0}\,Y^{3/2}, \tag{14}$$

$$\text{Newtonian } (Y \gg a_0^2/c^4): \quad J(Y) \to 0, \quad J'(Y) \to 0. \tag{15}$$

The transition region $Y \sim a_0^2/c^4$ is *unconstrained by the theory*. Different smooth interpolations between (**??**) and (**??**) correspond to different MOND interpolating functions $\mu(x)$.

This freedom has been present since Bekenstein and Milgrom [**?**] introduced AQUAL in 1984. Neither the τ framework nor the BS theory uniquely determines $J(Y)$. The τ framework does, however, suggest a specific form (Sec. **??**).

B. MOND phenomenology from $J(Y)$

[KNOWN] The deep-MOND limit of (**??**) produces the modified Poisson equation [**?** **?**]:

$$\boldsymbol{\nabla}\cdot\left[\mu\left(\frac{|\boldsymbol{\nabla}\Phi|}{a_0}\right)\boldsymbol{\nabla}\Phi\right] = 4\pi G\,\rho_{\rm b}, \tag{16}$$

where $\mu(x) \to 1$ for $x \gg 1$ (Newtonian) and $\mu(x) \to x$ for $x \ll 1$ (deep-MOND).

From (**??**) in the deep-MOND regime, the Baryonic Tully–Fisher Relation (BTFR) follows exactly [**?** **?**]:

$$V_{\rm flat}^4 = G\,M_{\rm bar}\,a_0, \tag{17}$$

and the Radial Acceleration Relation (RAR) [**?** **?**]:

$$g_{\rm obs} = \frac{g_{\rm bar}}{1-\exp\left(-\sqrt{g_{\rm bar}/a_0}\right)} \tag{18}$$

with $a_0 = (1.20{\pm}0.02){\times}10^{-10}$ m/s^2 and observed scatter 0.13 dex over 2693 data points in 153 SPARC galaxies [**?**].

C. The Crooks interpolation function

[NEW SYNTHESIS] The τ framework *suggests* a specific MOND interpolating function *inspired by* the Crooks fluctuation theorem [**?**]. For a process with entropy production Σ, the Crooks relation $P(\Sigma)/P(-\Sigma) = e^{\Sigma}$ motivates the ansatz

$$\mu(x) = 1 - e^{-\sqrt{x}}, \tag{19}$$

where $x = g_{\rm bar}/a_0$. The precise mapping from Crooks statistics to the MOND interpolation requires identifying Σ with $\sqrt{g_{\rm bar}/a_0}$, which remains a conjecture; the functional form is proposed as an ansatz whose deep-MOND

and Newtonian limits are correct by construction. Note that this $\mu(x)$ is the RAR interpolating function (as in Eq. (**??**)), not the AQUAL μ-function of Eq. (**??**); the two are related by $\nu(y) = 1/\mu(y)$ where ν is the RAR boosting function. Equation (**??**) satisfies $\mu(x) \to 1$ for $x \to \infty$ and $\mu(x) \to \sqrt{x}$ for $x \to 0$ (giving $V_f^4 = GM_{\rm bar}\, a_0$ in deep-MOND).

If the Crooks ansatz is validated against the SPARC 175-galaxy sample, this would eliminate the free function $J(Y)$ entirely, making the theory's galactic predictions parameter-free (given a_0). This is the single most impactful testable prediction of the τ framework for galactic phenomenology.

D. a_0 and μ_0: suggestive coincidence

The numerical relation

$$a_0 = \frac{c^2\mu_0}{2\pi} = \frac{cH_0}{2\pi} = 1.04 \times 10^{-10}\ \mathrm{m/s^2} \qquad (20)$$

(for $H_0 = 67.4$ km/s/Mpc) is within 13% of the empirical a_0. This "Milgrom coincidence" [**?**] has been known since the 1980s but has never been rigorously derived.

In the BS action, a_0 enters through $J(Y)$ and μ_0 enters through $K(Q)$. These are **structurally independent sectors** [**?**]: J depends on the spatial acceleration Y, while K depends on the temporal gradient Q. The two sectors decouple in the regimes where each is important (galactic for J, cosmological for K).

The connection (**??**) *is suggestive but unproven.* It may hint at a deeper unification of the J and K sectors, but no theoretical mechanism within the BS theory or the τ framework connects them.

V. OBSERVATIONAL TESTS

A. Low-redshift tests (11 criteria)

We systematically confront the theory with 11 low-redshift observational tests. The key distinction is that $w \neq c_s^2$ in the Khronon: w affects only the background expansion (Friedmann equation), while $c_s^2 = 0$ controls perturbations.

Table **??** summarizes the results. Ten tests are safely passed and one (SN Ia) is marginal at the 1σ level of the Pantheon+ uncertainty [**?**]. †The S_8 estimate is qualitative: $w > 0$ causes faster dilution of DM at late times, reducing structure growth; a full Boltzmann code computation is needed for the precise value. The TKS bound $c_s^2 < 3.21 \times 10^{-6}$ (99.7% CL) [**?**] is trivially satisfied: the Khronon-Tensor theory gives $c_s^2 = 0$ exactly (Sec. **??**).

The most striking result is the DESI BAO preference: $w(z{=}0) = 0.145$ reduces Ω_Λ by $\sim 5\%$, yielding an effective $w_{\rm DE} \sim -0.95$ (quintessence-like), which DESI data prefer at $\Delta\chi^2 = -3.9$ relative to ΛCDM.

TABLE II. Low-redshift observational tests. ✓ = safe; ∼ = marginal; × = tension. Quantitative entries ($\Delta\chi^2$, $A_{\rm ISW}$, σ_8) are author estimates from the GDM mapping of w into standard cosmological codes; a full Boltzmann solver analysis is pending.

Test	Result	Key quantity
Jeans length	✓	$\lambda_J = 0$ ($c_s^2 = 0$)
SN Ia distance	∼	$\lvert\delta\mu\rvert \sim 0.016$ mag
BAO / DESI	✓	$\Delta\chi^2 = -3.9$ (preferred)
Growth rate $f\sigma_8$	✓	$\Delta\chi^2 = +0.15$
S_8 tension	✓	$S_8 \lesssim 0.84$ (reduced†)
ISW effect	✓	$A_{\rm ISW} = 1.001$
TKS 2016 CMB	✓	$c_s^2 = 0$ exactly (no-go → tensor)
Weak lensing	✓	$< 1\%$ change
Peculiar velocities	✓	$< 1\%$ change
Cluster counts	✓	$\sigma_8 = 0.817$
BTFR scatter	✓	< 0.15 dex

B. Galactic-scale tests

The MOND sector $J(Y)$ has been tested extensively in the literature [**?** **?** **?**]. Key results relevant to the BS Khronon:

1. **BTFR**: $V_f^4 = G\,M_{\rm bar}\,a_0$ with observed scatter 0.10–0.15 dex across 10^7–$10^{12}\ M_\odot$ [**?** **?**];
2. **RAR**: 0.13 dex scatter over 2693 points in 153 galaxies [**?**] (measurement floor: ∼0.12 dex);
3. **External Field Effect**: detected at $> 5\sigma$ by Chae et al. [**?**]—a unique MOND prediction that ΛCDM cannot produce;
4. **Dwarf spheroidals**: Ghari & Haghi [**?**] found Verlinde's emergent gravity (sharing the same information-theoretic structure) preferred over MOND at 5.2σ for 23 dSphs;
5. **SPARC RAR scatter**: 0.144 dex (measurement limit 0.119 dex), a_0 best-fit 1.025×10^{-10} m/s^2; Gubitosi et al. [**?**] test RG-improved gravity on SPARC and find comparable fits;
6. **Wide binaries**: recent data from Gaia DR3 show 2–3σ hints of MOND-like behavior at low accelerations [**?**], though interpretation is debated.

C. Power spectrum and Fagin comparison

[KNOWN] At linear scales ($k < 0.1\ h$/Mpc), the Khronon power spectrum is *identical* to CDM because $c_s^2 = 0$. Skordis and Zlósnik [**?**] demonstrated that the closely related AeST theory reproduces both the CMB and $P(k)$.

At nonlinear scales, the MOND sector introduces deviations. Fagin et al. [?] measured the galaxy power spectrum slope $\beta = 5.22 \pm 0.41$ from SLACS strong lenses. The Khronon with isothermal approximation predicts $\beta \approx 6.2$ (2.4σ tension), while CDM is excluded at 6.8σ. The Khronon is the *closest* theoretical match, though the 2.4σ gap may indicate the need for a more complete Khronon rotation curve model beyond the isothermal approximation.

D. BBN, GW, and solar system

- **BBN**: the Khronon energy density at $z \sim 10^9$ is negligible ($w \to 0$, $\rho_K \to 0$). No modification to BBN;
- **GW170817**: $c_{\rm T} = c$ exactly ($c_{13} = 0$ by construction);
- **PPN**: $\beta_{\rm PPN} = \gamma_{\rm PPN} = 1$ (identical to GR at 1PN) [? ?];
- **Binary pulsars**: no dipolar radiation ($c_{13} = 0$) [?].

VI. WHAT THIS FRAMEWORK PREDICTS AND WHAT IT ASSUMES

A. Derived results

D1. $K(Q) = \mu^2(Q-1)^2$ from ghost condensation (leading-order);

D2. $c_{\rm s}^2 = 0$ from $K'(Q_0 = 1) = 0$ (radiatively stable);

D3. $c_{\rm T} = c$ from action choice ($c_{13} = 0$ by construction);

D4. $\Omega_{\rm DM} = \delta_0(2+\delta_0)/3 = 0.265$ (from $\mu_0 = H_0/c$ and Eq. (**??**));

D5. BTFR and RAR from $J(Y)$ (given the MOND interpolating function and a_0);

D6. Full stability (8 criteria, Sec. **??**).

B. Assumptions (not derived)

A1. $\mu_0 = H_0/c$: uniquely selected by dimensional analysis without Planck physics, but this is a naturality argument, not a derivation from field equations;

A2. Running $\mu(k) = k$ (or $\mu_{\rm bg} = H(z)/c$): well-motivated by 3 convergent arguments (Sec. **??**), but not a derivation. This is necessary for CMB compatibility;

TABLE III. Key falsifiable predictions. The "fatal?" column indicates whether failure would rule out the entire framework.

Prediction	**Timeline**	**Fatal?**
$w(z)$ evolution	DESI DR2 (2027)	Yes
DM particle detection	Ongoing	Yes
$c_{\rm s}^2 = 0$ (21cm + CMB)	2028–2030	Yes
$c_{\rm T} = c$ (exactly)	Ongoing (GW)	Yes
BH shadow 2.72 vs 2.60 r_s	EHT, 2030s	No*
a_0 universality (BIG-SPARC)	2026–2027	Partial
Crooks $\mu(x)$ vs SPARC	Near-term	No*
$w(z{=}0) = 0.145$	DESI/Euclid	Yes
No dipolar GW radiation	LISA (2030s)	Yes
EFE in wide binaries	Gaia DR4	Partial

A3. $J(Y)$ functional form: free function, only asymptotic behavior determined. The Crooks interpolation (**??**) is an ansatz inspired by thermodynamic fluctuation relations; a rigorous derivation from Crooks theorem remains open;

A4. a_0–μ_0 connection ($a_0 = cH_0/(2\pi)$ to 13%): suggestive but unproven;

A5. $c_{13} = 0$: action choice (not geometric necessity);

A6. $\Omega_{\rm DM}h^2 = 0.12$: initial condition, not predicted;

A7. Decoupling: the galactic sector $J(Y)$ and cosmological sector $K(Q)$ have no cross-coupling terms in the action. This is a simplification; cross-terms may be needed for a complete theory.

C. Falsifiable predictions

We identify specific, falsifiable predictions; Table **??** lists the ten most decisive:

The most stringent near-term test is the late-time expansion history: $w_0 = 0.145$ modifies the Hubble parameter at $z < 2$ relative to pure CDM. DESI DR2 and Euclid will directly probe this. Detection of a dark matter particle at $> 5\sigma$ would rule out the framework entirely.

*These tests would falsify specific sub-predictions but not the overall two-sector structure.

VII. CMB PREDICTIONS: WHY ΛCDM WORKS

A. The pure Khronon no-go theorem

We prove that a pure scalar Khronon field with ghost condensation cannot simultaneously produce the observed dark matter abundance *and* satisfy CMB constraints on the dark matter sound speed.

Theorem 2 (No-go for pure Khronon)**.** *For any analytic kinetic function $K(Q)$ with ghost condensation $K'(Q_0) = 0$, it is impossible to simultaneously achieve:*

(i) $\Omega_{\rm DM} = 0.265 \pm 0.005$,

(ii) $c_{\rm s}^2 < 10^{-6}$ *at recombination,*

(iii) CDM-like dilution $\rho_K \propto a^{-3}$.

Proof sketch. From the field equation $\partial_\mu(a^3 K'(Q)/N) = 0$, the deviation $\delta = Q/Q_0 - 1$ evolves as $\delta \propto a^{-3}$ in the matter era. The effective sound speed is $c_{\rm s}^2 = \delta/(2+3\delta)$ for small δ, saturating at $c_{\rm s}^2 = 1/3$ for $\delta \gg 1$.

Requiring $\Omega_{\rm DM} = 0.265$ fixes $\delta(a_{\rm rec}) \sim 10^9$ (far from the small-δ regime), giving $c_{\rm s}^2(a_{\rm rec}) \to 1/3$, which is excluded by Planck at $> 55\sigma$ (Thomas, Kopp & Skordis [**?**]). No choice of $K(Q)$ avoids this: the saturation is structural.

Completeness.—The value $\delta(a_{\rm rec})$ is a non-negative real number, so exactly three regimes are possible: $\delta \ll 1$, $\delta \gg 1$, or $\delta \sim O(1)$. In the first case the a^{-3} dilution requirement forces $\delta(a_{\rm rec})$ too small for the observed $\Omega_{\rm DM}$; in the second and third cases $c_{\rm s}^2 \geq O(0.1)$, violating condition (ii). No fine-tuned $\mu(a)$ can avoid this trichotomy: regardless of the path $\mu(a)$ takes, $\delta(a_{\rm rec})$ must fall in one of these three regimes, and each violates at least one condition. See Supplemental Material for the complete proof. □

B. Resolution: the Khronon-Tensor theory

Blanchet & Skordis [**?**] showed that adding a symmetric tensor field $B_{\mu\nu}$ to the Khronon action resolves the no-go theorem. The tensor field absorbs the DBI pressure, yielding $c_{\rm s}^2 = 0$ *exactly* at the perturbation level while preserving $\rho \propto a^{-3}$ at the background level.

This is not an ad hoc fix: the tensor field is the natural completion of the Khronon field theory, analogous to how the photon field completes the electron field in QED.

C. Prediction: CMB = ΛCDM

With the Khronon-Tensor theory:

- The background evolution is *identical* to ΛCDM ($\rho_K \propto a^{-3}$, $\Omega_K = \Omega_{\rm cdm}$).
- The perturbation sound speed is $c_{\rm s}^2 = 0$ *exactly*, not approximately.
- The CMB power spectrum C_ℓ^{TT} is *identical* to ΛCDM at all multipoles.

This is *consistent* with Planck data [**?**], which shows no deviation from ΛCDM below the percent level. The theory *predicts* this agreement — it is not imposed by hand.

D. Why this is significant

The $\Sigma = 2\ln Q$ framework provides a unified explanation for two seemingly contradictory observations:

1. **CMB = ΛCDM**: because the no-go theorem forces $c_{\rm s}^2 = 0$ (via the tensor field), making Khronon dark matter indistinguishable from CDM at linear perturbation scales.
2. **Galaxies = MOND**: because the DBI kinetic structure $J(Y)$ produces the MOND interpolating function $\mu(x)$ via the Crooks fluctuation theorem (Sec. **??**).

The transition between these two regimes occurs at the MOND acceleration scale $a_0 = 2\pi c^2(1+\Omega_b)/[r_d(1+z_{\rm dec})]$, which is *derived* (0.3% accuracy), not fitted.

VIII. DISCUSSION

A. Comparison with AeST

The BS Khronon and AeST [**?** **?** **?** **?**] are closely related: both are relativistic MOND theories with a scalar field whose cosmological perturbations have $c_{\rm s}^2 = 0$. Key differences:

*The no-go theorem (Sec. **??**) forces the Khronon-Tensor completion [**?**], which gives $c_{\rm s}^2 = 0$ exactly and C_ℓ^{TT} identical to ΛCDM at all multipoles.

B. The two-sector structure: feature or bug?

The structural independence of $J(Y)$ and $K(Q)$ is arguably both the theory's greatest strength and its most significant gap:

Strength: Each sector can be independently tested and constrained. The cosmological sector is fully specified (1 parameter), while the galactic sector inherits the extensive MOND phenomenological success.

Gap: There is no known theoretical principle connecting J to K. The coincidence $a_0 \sim c^2\mu_0/(2\pi)$ hints at

TABLE IV. BS Khronon vs. AeST comparison.

Feature	**BS Khronon**	**AeST**
Fields	$g_{\mu\nu} + \phi$	$g_{\mu\nu} + A^\mu + \phi$
DoF	3 (2T + 1S)	6 (2T + 2V + 2S)
Free function	$J(Y)$ (MOND only)	$F(Y, Q)$ (combined)
$K(Q)$ fixed?	Yes (quadratic)	Part of F
$c_{\rm T} = c$	By construction	By construction
Theory-specific params	∼3	∼5
τ interpretation	Yes	Not yet
CMB fit demonstrated?	$c_{\rm s}^2 = 0$ exactly*	Yes [**?**]

a connection, but it may require a deeper theory (perhaps the unified $\Sigma = D(\rho_{\rm spacetime}\|\rho_{\rm matter})$ proposed in Paper 4 [?]) to derive it.

In AeST, the analogous gap is papered over by using a single free function $F(Y, Q)$—but the Y-dependence and Q-dependence within F remain effectively independent. The BS approach is more honest: two separate functions for two separate physical regimes.

C. What this framework does *not* explain

1. **Why $\Omega_{\rm DM}h^2 = 0.12$**: this is an initial condition (I_0), not predicted;

2. **Why $J(Y)$ takes a particular form**: the transition region is unconstrained unless the Crooks interpolation (**??**) is validated;

3. **Why μ runs as k^1**: this is assumed, not derived from field equations;

4. **Cluster-scale mass discrepancies**: see the Bullet Cluster discussion below;

5. **The $(2+\delta)$ redshift correction** (Sec. **??**): the factor $(2+\delta(z))$ in ρ_K introduces a $\sim$20% variation at recombination relative to pure a^{-3} scaling.

D. Bullet Cluster

The Bullet Cluster (1E 0657-558) presents a well-known challenge for modified gravity theories: the gravitational lensing mass is spatially offset from the baryonic gas [?]. In the Khronon-Tensor framework, this is naturally explained: the no-go theorem (Theorem **??**) forces $c_{\rm s}^2 = 0$ *exactly* via the tensor field completion, meaning the Khronon condensate behaves as *pressureless dust* during cluster collisions. Like CDM particles, the Khronon energy density passes through the collision region without interacting with the gas, reproducing the observed mass-gas separation.

This represents a significant improvement over traditional MOND theories, which struggle with the Bullet Cluster because they lack a collisionless dark component. The Khronon condensate *is* that component, with its collisionless nature guaranteed by the Khronon-Tensor structure that also ensures CDM-like perturbation growth in the CMB (Sec. **??**).

E. Connection to Papers 1–2 and outlook

The logical chain across papers is:

$$\begin{aligned}
&\text{Paper 1: } \tau = 1 - F,\ \ \Sigma = D - D \circ \mathcal{N},\\
&\text{Paper 2: } \Sigma_{\rm grav} = -\ln(-g_{00}),\\
&\text{Paper 3: } K(Q),\ J(Y) \ \to\ \text{DM + MOND},\\
&\text{Paper 4: } \Sigma = D(\rho_{\rm st}\|\rho_{\rm m}) \ \to\ \text{unification.}
\end{aligned}$$

The bridge between Papers 2 and 3 is provided by:

Theorem 3 (Gravitational Entropy Production)**.** *Let (M, g_{ab}) be a globally hyperbolic spacetime foliated by the Khronon field ϕ with inverse lapse $Q = 1/N_\phi = \sqrt{-g^{ab}\nabla_a\phi\,\nabla_b\phi}$. For a probe mode of proper frequency ω, the entropy production of the gravitational channel satisfies*

$$\Sigma = 2\ln Q + O(\varepsilon_{\rm NA}^2), \tag{21}$$

where $\varepsilon_{\rm NA} = |d\ln Q/d\tau|/(2\omega)$. In particular, $\Sigma = 2\ln Q$ exactly on stationary backgrounds ($\varepsilon_{\rm NA} = 0$) and $\Sigma \geq 2\ln Q$ on all backgrounds.

Proof sketch. In Khronon-adapted coordinates, the ADM decomposition gives $ds^2 = -Q^{-2}\,d\phi^2 + h_{ij}(dx^i + N^i d\phi)(dx^j + N^j d\phi)$. A WKB probe mode has proper frequency $\omega_{\rm proper} = \omega_\phi/Q$. The intensity transmissivity of the gravitational channel is $\eta = (\omega_{\rm out}/\omega_{\rm in})^2 \times (dt_{\rm out}/dt_{\rm in})^{-1} = 1/Q^2$ (both energy per quantum and rate of quanta redshift by $1/Q$). By the Ivan–Sabapathy–Simon result for bosonic Gaussian channels, $\Sigma = -\ln\eta = 2\ln Q$. The Bogoliubov first correction gives $|beta/\alpha|^2 \leq \varepsilon_{\rm NA}^2$, which is non-negative (particle creation adds entropy), yielding the inequality $\Sigma \geq 2\ln Q$. □

This theorem unifies the results of Papers 2 and 3: Paper 2's $\Sigma_{\rm grav} = -\ln(-g_{00})$ is the static limit ($Q = 1/\sqrt{-g_{00}}$), while the cosmological sector's $\Sigma_{\rm cosmo} = 2\ln(1+\delta)$ is the FRW limit ($Q_0 = 1+\delta$).

Papers 1–3 establish the τ framework at three scales (quantum information, strong field, weak field). Paper 4 [?] will argue these are three limits of a single equation $\Sigma = D(\rho_{\rm spacetime}\|\rho_{\rm matter})$ under different boundary conditions. The key open problems for this unification are: (i) deriving the running $\mu(k) = k$ from first principles; (ii) connecting $J(Y)$ to $K(Q)$ through a unified variational principle; and (iii) determining whether the Crooks interpolation function is the unique form selected by the theory.

The role of the no-go theorem.—The no-go theorem (Theorem **??**) is not a weakness of the Khronon framework but a *structural prediction*: it tells us that the pure scalar theory is incomplete and that the tensor field completion [?] is *necessary*. This is analogous to how the inconsistency of scalar QED at loop level necessitates the full gauge theory. The resulting Khronon-Tensor theory gives $c_{\rm s}^2 = 0$ exactly, which is *stronger* than a parametric suppression—it is a structural zero, not an accidental cancellation.

IX. CONCLUSION

We have presented the weak-field phenomenology of the τ framework, realized through the Blanchet–Skordis Khronon action. The theory has two independent sectors:

1. The **cosmological sector** $K(Q) = \mu^2(Q-1)^2$ is fully determined by ghost condensation with $\mu_0 = H_0/c$ from dimensional analysis. It reproduces the cosmological dark matter density and passes 10 of 11 low-redshift observational tests with one marginal (assuming running $\mu_{\rm bg} = H/c$). A no-go theorem (Theorem **??**) establishes that the pure scalar Khronon is incompatible with CMB observations, necessitating the Blanchet–Skordis tensor field completion [**?**]. The resulting Khronon-Tensor theory gives $c_{\rm s}^2 = 0$ *exactly*, predicting a CMB power spectrum identical to ΛCDM. The Bullet Cluster mass-gas offset is naturally explained by the collisionless ($c_{\rm s}^2 = 0$) Khronon condensate.

2. The **galactic sector** $J(Y)$ reproduces MOND phenomenology (BTFR, RAR, EFE) with one empirical parameter a_0 and one free function. The Crooks interpolation function provides a specific, testable prediction.

The $\Sigma = 2\ln Q$ framework naturally explains why ΛCDM succeeds at cosmological scales (the no-go theorem forces $c_{\rm s}^2 = 0$) while galaxies follow MOND (the DBI kinetic structure produces $\mu(x)$ via Crooks). The transition scale a_0 is derived to 0.3% accuracy from recombination physics.

The theory is structurally stable (no ghosts, no gradient instabilities, no tachyons), satisfies $c_{\rm T} = c$ exactly, and has fewer free parameters than AeST.

We have been explicit about what is derived (ghost condensation, $c_{\rm s}^2 = 0$, the no-go theorem, stability), what is assumed ($\mu_0 = H_0/c$, running μ, $c_{13} = 0$, a_0–μ_0 connection), and what is open ($J(Y)$ form, $\Omega_{\rm DM}h^2$ value, microscopic running derivation). Key falsifiable predictions are identified, with the $w(z)$ evolution curve as the most decisive near-term test.

The dark matter problem may not require new particles. It may instead reflect the thermodynamic cost of temporal ordering—the price we pay for having a well-defined arrow of time in a universe with a cosmological horizon.

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